Lumina AI Study Coach

Sample Study Material • Computer Science

Topic 1: Artificial Intelligence

Meaning: Artificial Intelligence (AI) is a field of computer science that develops systems able to perform tasks that usually require human intelligence, such as learning, problem-solving, and understanding language.

Examples: voice assistants, recommendation systems, image recognition, and chatbots.

Topic 2: Machine Learning

Meaning: Machine Learning (ML) is a part of AI in which computers learn patterns from data and use those patterns to make predictions or decisions.

Main types:

Type	Simple explanation	Example
Supervised learning	Learns from labelled examples.	Spam email detection
Unsupervised learning	Finds patterns in unlabelled data.	Grouping similar customers
Reinforcement learning	Learns through rewards and penalties.	Training a game-playing agent

Topic 3: Computer Vision

Meaning: Computer Vision helps computers interpret information from images and videos.

Common applications: face detection, medical image analysis, object tracking, and image search.

Quick Revision Questions

1. Define Artificial Intelligence.
2. What is Machine Learning?
3. Name the three main types of Machine Learning.
4. Write any two applications of Computer Vision.

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